

capital and infrastructure barrier

1. High chip fabrication setup cost
2. inadequate semiconductor manufacturing infrastructure
3. inconsistent power and water supply
4. Road blocks in EDA tool manufacturing
5. underdeveloped experimental infrastructure

Regulatory Compliance barriers

1. slow pace of indian semiconductor mission (ISM & SPECS)
2. complex approval process for industrial collaboration (limited subsidies / lack of long term policy support / complex approval process / uncertainty over intellectual property rights)
3. high and complex tax regimes (import & export)
4. inconsistent product & design incentives (PLI & DLI)
5. unresolved import dependency due to vague ITA rules (MSIP & ITA-1 initiatives)

Supply Chain barriers

1. globally fragmented semiconductor supply chain management
2. inadequate & fragile domestic semiconductor ecosystem
3. lack of value chain integration for semiconductor development
4. lack of specific and sophisticated transportation network
5. globally fragmented chip supply chain crisis

workforce barrier

1. intellectual capital loss
2. competency & talent gap
3. industry academia alignment issue
4. constrained knowledge exchange

Global Conflicts & Trade barriers	Research and Technological barriers	Environment and sustainability barriers
1. oligopoly in semiconductor industry geopolitical power struggle (tension/dominance)	1. restricted access to cutting edge technology and materials	1. resource intensity
2. US China trade conflicts & restrictions	2. IP management & data security challenges	2. environmental regulatory challenges
3. Taiwan dominance and alliances	3. limited scalability and efficiency in semiconductor production	3. legal impediments due to SDG- 6 (clean water & sanitation)
4. global trade tension & instability in SSCM	4. constrained semiconductor logic design ecosystem	4. SDG 12 /(Alag karo) inadequate recycling and waste management (
5. Global currency volatility drives price instability in SSCM	5. oligopoly in electronics chips demand	5. stringest environment regulation (water act 1974/ underground water act 2022/ environment protection act 1986/ management and transboundary movement rules 2016) (Combine , Compress & Rename in Indian Context)
6. global market instability and subsidy tensions		